

Exercises

Level - 1

(Problems Based on Fundamentals)

Direct Substitution Method (DSM)

1. Evaluate : $\lim_{x \rightarrow 1} (x^2 - 6x + 10)$

2. Evaluate : $\lim_{x \rightarrow 1} (x^{2016} - x^{2017} + 2)$

Factorisation Method (FM)

3. Evaluate : $\lim_{x \rightarrow 1} \left(\frac{x^2 - 3x + 2}{x^2 - 5x + 4} \right)$

4. Evaluate : $\lim_{x \rightarrow 1} \left(\frac{1 - x^{\frac{1}{3}}}{1 - x^{\frac{2}{3}}} \right)$

5. Evaluate : $\lim_{x \rightarrow 2} \left(\frac{x^3 + 7x^2 - 36}{x^2 + 2x - 8} \right)$

6. Evaluate : $\lim_{x \rightarrow 1} \left(\frac{x^2 + x \log x - \log x - 1}{x^3 - 1} \right)$

Rationalisation Method (RM)

7. Evaluate : $\lim_{x \rightarrow 0} \frac{x}{\sqrt{a+x} - \sqrt{a-x}}$

8. Evaluate : $\lim_{x \rightarrow 0} \left(\frac{\sqrt{1+x} - 1}{x} \right)$

9. Evaluate : $\lim_{x \rightarrow 0} \frac{3 - \sqrt{5+x}}{1 - \sqrt{5-x}}$

10. Evaluate : $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{2x}$

11. Evaluate : $\lim_{x \rightarrow 0} \frac{\sqrt{1+4x} - \sqrt{5+2x}}{x-2}$

12. Evaluate : $\lim_{x \rightarrow \sqrt{10}} \left(\frac{\sqrt{7+2x} - \sqrt{5} - \sqrt{2}}{x^2 - 10} \right)$

13. Evaluate : $\lim_{x \rightarrow a} \frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}}$

14. Evaluate : $\lim_{x \rightarrow -1} \frac{\sqrt[3]{7-x} - 2}{(x+1)}$

15. Evaluate : $\lim_{x \rightarrow 1} \left(\frac{\sqrt{x^2+8} - \sqrt{10-x^2}}{\sqrt{x^2+3} - \sqrt{5-x^2}} \right)$

Standard Result Method (SRM)

16. Evaluate : $\lim_{x \rightarrow a} \frac{x\sqrt{x} - a\sqrt{a}}{x-a}$

17. Evaluate : $\lim_{x \rightarrow a} \frac{(x+2)^{\frac{5}{3}} - (a+2)^{\frac{5}{3}}}{x-a}$

18. Evaluate : $\lim_{x \rightarrow a} \frac{x^m - a^m}{x^n - a^n}$

19. Evaluate : $\lim_{x \rightarrow 0} \frac{(1-x)^n - 1}{x}$

20. Evaluate: $\lim_{x \rightarrow 1} \frac{(x + x^2 + x^3 + \dots + x^n) - n}{x - 1}$

21. Evaluate: $\lim_{x \rightarrow 1} \frac{x^{n+1} - (n+1)x + n}{(x-1)^2}$

22. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2} \right)$

23. Evaluate: $\lim_{x \rightarrow 1} \left(\frac{\sqrt[3]{7+x^3} - \sqrt[2]{3+x^2}}{x-1} \right)$

24. Evaluate: $\lim_{x \rightarrow 1} \left(\frac{\sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} - 3}{x-1} \right)$

25. Evaluate: $\lim_{x \rightarrow 1} \left(\frac{\left(\sum_{k=1}^{100} x^k \right) - 100}{x-1} \right)$

Infinity Method (IM)

26. Evaluate: $\lim_{n \rightarrow \infty} (4^n + 5^n)^{\frac{1}{n}}$

27. Evaluate: $\lim_{n \rightarrow \infty} \left(\frac{n^2}{1+2+3+\dots+n} \right)$

28. Evaluate: $\lim_{n \rightarrow \infty} \left(\frac{x^n + y^n}{x^n - y^n} \right)$, where $0 < x < y$.

29. Evaluate: $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 2x} - x)$

30. Evaluate: $\lim_{x \rightarrow -\infty} (\sqrt{x^2 + 8x} + x)$

31. Evaluate: $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 2x - 1}{2x^2 - 3x - 2} \right)^{\frac{2x+5}{2x-7}}$

32. Evaluate:

$$\lim_{x \rightarrow \infty} \left(\frac{(x+1)^{10} + (x+2)^{10} + \dots + (x+2012)^{10}}{x^{10} + 2012^{10}} \right)$$

33. Evaluate:

$$\lim_{n \rightarrow \infty} \left(\frac{1 \cdot n + 2 \cdot (n-1) + 3 \cdot (n-2) + \dots + n \cdot 1}{n^3} \right)$$

34. Evaluate:

$$\lim_{n \rightarrow \infty} n^2 \left\{ \sqrt{\left(1 - \cos \frac{1}{n}\right)} \sqrt{\left(1 - \cos \frac{1}{n}\right)} \sqrt{\left(1 - \cos \frac{1}{n}\right)} \dots \infty \right\}$$

35. If $\lim_{x \rightarrow \infty} \left(\frac{x^2 - 1}{x + 1} - ax - b \right) = 2$, find the values of a and b .

36. If $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 1}{x + 1} - ax - b \right) = 0$, find the values of a and b .

37. If $\lim_{x \rightarrow 0} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$, then find the values of a and b .

38. Evaluate $\lim_{x \rightarrow -\infty} \left(\frac{x^4 \sin\left(\frac{1}{x}\right) + x^2}{1 + |x|^3} \right)$

Trigonometric Limit

39. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\sin 3x}{5x} \right)$

40. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\sin ax}{\sin bx} \right)$

41. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\sin 2x + 3x}{4x + \sin 6x} \right)$

42. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\sin 2x + \sin 5x}{\sin 4x + \sin 6x} \right)$

43. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{1 - \cos 4x}{1 - \cos 6x} \right)$

44. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\sqrt{1 - \cos 2x}}{x^2} \right)$

45. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\cos 7x - \cos 9x}{\cos 3x - \cos 5x} \right)$

46. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{1 - \cos x \sqrt{\cos 2x}}{x^2} \right)$

47. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\tan x - \sin x}{x^3} \right)$

48. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{1 - \cos x \cos 2x \cos 3x}{\sin^2 2x} \right)$

49. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\sin^{-1} x - \tan^{-1} x}{x^3} \right)$

50. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{x - \sin x}{x^3} \right)$

51. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{x - \tan x}{x^3} \right)$

52. Evaluate: $\lim_{x \rightarrow \infty} \left(\frac{x + \sin x}{x + \cos x} \right)$

53. Evaluate: $\lim_{x \rightarrow 0} \left(\left[\frac{n \sin x}{x} \right] + \left[\frac{n \tan x}{x} \right] \right)$

54. If $\min(x^2 + 2x + 3, x^2 + 4x + 10)$

and $b = \lim_{x \rightarrow 0} \left(\frac{1 - \cos \theta}{\theta^2} \right)$, then find the value

of $\sum_{r=0}^n (a^r b^{n-r})$.

55. If $\lim_{x \rightarrow 0} \left(\frac{a \cos x + bx \sin x - 5}{x^4} \right)$ exists and finite, then

find the value of $a + 2b + 10$.

56. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\sin(\pi \cos^2 x)}{x^2} \right)$

Exponential Limit

57. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{e^{4x} - 1}{5x} \right)$

58. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{e^{3x} - 1}{e^{5x} - 1} \right)$

59. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{e^x + e^{-x} - 2}{x^2} \right)$

60. Evaluate: $\lim_{x \rightarrow 0} \frac{e^{3+x} - \sin x - e^3}{x}$

61. Evaluate: $\lim_{x \rightarrow 0} \frac{e^x - e^{\sin x}}{x - \sin x}$

62. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{e^x - e^{x \cos x}}{x + \sin x} \right)$

63. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{e^x - 1 - x}{x^2} \right)$

64. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{8^x - 4^x - 2^x + 1}{x^2} \right)$

65. Evaluate: $\lim_{x \rightarrow 0} \frac{9^x - 2 \cdot 6^x + 4^x}{x^2}$

66. Evaluate: $\lim_{x \rightarrow a} \left(\frac{a^x - a^a}{x - a} \right), a > 0$

67. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{a^{x+h} + a^{x-h} - 2a^x}{h^2} \right)$

68. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{8^x - 7^x}{6^x - 5^x} \right)$

69. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{(5^x - 1)(4^x - 1)}{(3^x - 1)(6^x - 1)} \right)$

70. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{x \cdot 2^x - x}{1 - \cos x} \right)$

71. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{e^x - e^{-x} - 2x}{x - \sin x} \right)$

72. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{e^{x^3} - 1 - x^3}{\sin^6 2x} \right)$

73. Evaluate: $\lim_{x \rightarrow 2} \left(\frac{(\cos \alpha)^x + (\sin \alpha)^x - 1}{x - 2} \right)$

74. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{e^{x \cos x} - 1 - x}{\sin(x^2)} \right)$

75. If $a = \min \{x^2 + 4x + 6, x^2 + 2x + 8\}$

and $b = \lim_{x \rightarrow 0} \left(\frac{\sin x \cos x}{e^x - e^{-x}} \right)$, then find the value of

$$\sum_{r=0}^n (a^r b^{n-r})$$

76. If $\lim_{x \rightarrow 0} \left(\frac{Ae^x - B \cos x + Ce^{-x}}{x \sin x} \right) = 2$ then find the value of $A + B + C + 10$.

Logarithmic Limit

77. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\log(1 + 3x)}{\sin 2x} \right)$

78. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\log(1 + 3x)}{\log(1 - 2x)} \right)$

79. Evaluate: $\lim_{x \rightarrow e} \left(\frac{\log x - 1}{x - e} \right)$

80. Evaluate: $\lim_{x \rightarrow 5} \left(\frac{\log x - \log 5}{x - 5} \right)$

81. Evaluate: $\lim_{x \rightarrow 5} \left(\frac{\log(x + 5) - \log(5 - x)}{x - 5} \right)$

82. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{e^x - \log(x + e)}{e^x - 1} \right)$

83. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\ln(\cos x)}{\sqrt[4]{1 + x^2} - 1} \right)$

84. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\ln(\cos(\sin x))}{x^2} \right)$

85. Evaluate: $\lim_{x \rightarrow \infty} \left(x - x^2 \ln \left(1 + \frac{1}{x} \right) \right)$

86. Evaluate: $\lim_{x \rightarrow 1} \left(\frac{x^x - 1}{x \ln x} \right)$

87. Evaluate: $\lim_{x \rightarrow 1} \left(\frac{1}{\ln x} - \frac{1}{x - 1} \right)$

88. Evaluate: $\lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{\ln \cot x}{1 - \cot x} \right)$

L'Hospital Rule

89. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{e^x - 1 - x}{x^2} \right)$

90. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{e^{x^3} - 1 - x^3}{64x^6} \right)$

91. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{e^x - e^{-x} - 2x}{x - \sin x} \right)$

92. Evaluate: $\lim_{x \rightarrow e} \left(\frac{\ln x - 1}{x - e} \right)$

93. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{x + \tan 2x}{x - \tan 2x} \right)$

94. Evaluate: $\lim_{x \rightarrow e} \left(\frac{\sin x - x}{x^3} \right)$

95. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{e^x + e^{-x} - x^2 - 2}{\sin^2 x - x^2} \right)$

96. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{xe^x - \log(1+x)}{x^2} \right)$

97. Evaluate: $\lim_{x \rightarrow \infty} \left(\frac{\ln x}{x} \right)$

98. Evaluate: $\lim_{x \rightarrow \infty} \left(\frac{x}{e^x} \right)$

99. Evaluate: $\lim_{x \rightarrow \infty} \left(\frac{x^{2014}}{e^x} \right)$

100. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{x^2 + 2\cos x - 2}{x \sin^3 x} \right)$

101. Evaluate: $\lim_{x \rightarrow \frac{\pi}{2}} (\tan x \log_e \sin x)$

Advanced Exponential Limit

102. Evaluate: $\lim_{x \rightarrow \infty} \left(\frac{x+6}{x+1} \right)^{x+4}$

103. Evaluate: $\lim_{x \rightarrow \infty} \left(\frac{2x+4}{2x+3} \right)^{2x+10}$

104. Evaluate: $\lim_{x \rightarrow \infty} \left(\sin\left(\frac{1}{x}\right) + \cos\left(\frac{1}{x}\right) \right)^x$

105. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x}{3} \right)$

106. Evaluate: $\lim_{x \rightarrow 1} \left(2 - \frac{1}{x} \right)^{\tan\left(\frac{\pi x}{2}\right)}$

107. Evaluate: $\lim_{x \rightarrow 1} \left(\tan\left(\frac{\pi}{4} + \ln x\right) \right)^{\frac{1}{\ln x}}$

108. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\ln\left(\tan\left(\frac{\pi}{4} + 2x\right)\right)}{\sin 3x} \right)$

109. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x}$

110. Evaluate: $\lim_{x \rightarrow 0} \left(\frac{1 + \tan x}{1 + \sin x} \right)^{\operatorname{cosec} x}$

111. Evaluate: $\lim_{n \rightarrow \infty} \left(\frac{a - 1 + \sqrt[n]{b}}{a} \right)^n$

Sandwich Theorem

112. Evaluate: $\lim_{x \rightarrow 0} \left(x^2 \sin\left(\frac{1}{x}\right) \right)$

113. Evaluate: $\lim_{x \rightarrow \infty} \left(\frac{\sin x}{x} \right)$

114. Evaluate: $\lim_{x \rightarrow \infty} \left(\frac{[x]}{x} \right)$

115. Evaluate: $\lim_{x \rightarrow \infty} \left(\frac{x}{[x]} \right)$

116. Evaluate: $\lim_{n \rightarrow \infty} \left(\frac{[x] + [2x] + [3x] + \dots + [nx]}{n^2} \right)$

Newton and Leibnitz Rule

117. Evaluate: $\lim_{x \rightarrow 0} \frac{\left(x - \int_0^x \cos(t^2) dt \right)}{x^3 - 6x}$

118. Evaluate: $\lim_{x \rightarrow \infty} \frac{\left(\int_0^x \sqrt{4+t^4} dt \right)}{x^3}$

119. Evaluate: $\lim_{x \rightarrow 1} \frac{\left(\int_x^{x^2} e^{-t^2} dt \right)}{x - 1}$

Definite Integral as the limit of a sum

120. Evaluate: $\lim_{n \rightarrow \infty} \left[\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n} \right]$

121. Evaluate: $\lim_{n \rightarrow \infty} \left[\frac{1}{n^2+1^2} + \frac{1}{n^2+2^2} + \frac{1}{n^2+3^2} + \dots + \frac{1}{2n} \right]$

122. Evaluate: $\lim_{n \rightarrow \infty} \sum_{r=1}^{2n} \left(\frac{r}{\sqrt{n^2 + r^2}} \right)$

123. Evaluate: $\lim_{n \rightarrow \infty} \sum_{r=1}^{2n} \frac{2}{n} \left(\frac{2r}{n} + 1 \right)$

124. Evaluate: $\lim_{n \rightarrow \infty} \left(\frac{(2n)!}{n! \times n^n} \right)^{1/n}$

Advanced limit in the form of $(0^0 \text{ and } \infty^0)$

125. Evaluate: $\lim_{x \rightarrow 0^+} (x - x^2)^{x^2}$

Hints & Solutions

Level - 1

$$\begin{aligned} 1. \quad \lim_{x \rightarrow 1} (x^2 - 6x + 10) \\ &= 1 - 6 + 10 \\ &= 11 - 6 \\ &= 5 \end{aligned}$$

$$\begin{aligned} 2. \quad \lim_{x \rightarrow 1} (x^{2016} - x^{2017} + 2) \\ &= 1 - 1 + 2 \\ &= 2 \end{aligned}$$

$$\begin{aligned} 3. \quad \lim_{x \rightarrow 1} \left(\frac{x^2 - 3x + 2}{x^2 - 5x + 4} \right) \\ &= \lim_{x \rightarrow 1} \left(\frac{(x-1)(x-2)}{(x-1)(x-4)} \right) \\ &= \lim_{x \rightarrow 1} \left(\frac{(x-2)}{(x-4)} \right) \\ &= \left(\frac{1-2}{1-4} \right) = \frac{-1}{-3} = \frac{1}{3} \end{aligned}$$

$$\begin{aligned} 4. \quad \lim_{x \rightarrow 1} \left(\frac{1 - x^{-\frac{1}{3}}}{1 - x^{-\frac{2}{3}}} \right) \\ &= \lim_{x \rightarrow 1} \left(\frac{1 - x^{-\frac{1}{3}}}{(1)^2 - \left(x^{-\frac{1}{3}} \right)^2} \right) \\ &= \lim_{x \rightarrow 1} \left(\frac{\left(1 - x^{-\frac{1}{3}} \right)}{\left(1 + x^{-\frac{1}{3}} \right) \left(1 - x^{-\frac{1}{3}} \right)} \right) \\ &= \lim_{x \rightarrow 1} \left(\frac{1}{\left(1 + x^{-\frac{1}{3}} \right)} \right) = \frac{1}{2} \end{aligned}$$

$$\begin{aligned}
 6. \lim_{x \rightarrow 1} \left(\frac{x^2 + x \log x - \log x - 1}{x^3 - 1} \right) \\
 &= \lim_{x \rightarrow 1} \left(\frac{(x^2 - 1) + (x - 1) \log x}{x^3 - 1} \right) \\
 &= \lim_{x \rightarrow 1} \left(\frac{(x + 1) + \log x}{(x^2 + x + 1)} \right) \\
 &= \frac{2}{3}
 \end{aligned}$$

$$\begin{aligned}
 7. \lim_{x \rightarrow 0} \left(\frac{x}{\sqrt{a+x} - \sqrt{a-x}} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{x}{\sqrt{a+x} - \sqrt{a-x}} \times \frac{\sqrt{a+x} + \sqrt{a-x}}{\sqrt{a+x} + \sqrt{a-x}} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{x(\sqrt{a+x} + \sqrt{a-x})}{(a+x - a+x)} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{x(\sqrt{a+x} + \sqrt{a-x})}{2x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{(\sqrt{a+x} + \sqrt{a-x})}{2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\sqrt{a} + \sqrt{a}}{2} \right) = \frac{2\sqrt{a}}{2} = \sqrt{a}
 \end{aligned}$$

$$\begin{aligned}
 8. \lim_{x \rightarrow 0} \left(\frac{\sqrt{1+x} - 1}{x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\sqrt{1+x} - 1}{x} \times \frac{\sqrt{1+x} + 1}{\sqrt{1+x} + 1} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{1+x-1}{x(\sqrt{1+x} + 1)} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{x}{x(\sqrt{1+x} + 1)} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{1}{(\sqrt{1+x} + 1)} \right) \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 9. \lim_{x \rightarrow 0} \frac{3 - \sqrt{5+x}}{1 - \sqrt{5-x}} \\
 &= \left(\frac{3 - \sqrt{5}}{1 - \sqrt{5}} \right)
 \end{aligned}$$

$$10. \lim_{x \rightarrow 0} \left(\frac{\sqrt{1+x} - \sqrt{1-x}}{2x} \right)$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{2x} \times \frac{\sqrt{1+x} + \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \\
 &= \lim_{x \rightarrow 0} \left(\frac{1+x-1+x}{2x(\sqrt{1+x} + \sqrt{1-x})} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{2x}{2x(\sqrt{1+x} + \sqrt{1-x})} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{1}{(\sqrt{1+x} + \sqrt{1-x})} \right) \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 11. \lim_{x \rightarrow 2} \left(\frac{\sqrt{1+4x} - \sqrt{5+2x}}{x-2} \right) \\
 &= \lim_{x \rightarrow 2} \left(\frac{\sqrt{1+4x} - \sqrt{5+2x}}{x-2} \times \frac{\sqrt{1+4x} + \sqrt{5+2x}}{\sqrt{1+4x} + \sqrt{5+2x}} \right) \\
 &= \lim_{x \rightarrow 2} \left(\frac{1+4x-5-2x}{(x-2)(\sqrt{1+4x} + \sqrt{5+2x})} \right) \\
 &= \lim_{x \rightarrow 2} \left(\frac{2x-4}{(x-2)(\sqrt{1+4x} + \sqrt{5+2x})} \right) \\
 &= \lim_{x \rightarrow 2} \left(\frac{2(x-2)}{(x-2)(\sqrt{1+4x} + \sqrt{5+2x})} \right) \\
 &= \lim_{x \rightarrow 2} \left(\frac{2}{\sqrt{1+4x} + \sqrt{5+2x}} \right) \\
 &= \frac{2}{\sqrt{9} + \sqrt{9}} = \frac{2}{6} = \frac{1}{3}
 \end{aligned}$$

$$\begin{aligned}
 12. \lim_{x \rightarrow \sqrt{10}} \left(\frac{\sqrt{7+2x} - (\sqrt{5} + \sqrt{2})}{x^2 - 10} \right) \\
 &= \lim_{x \rightarrow \sqrt{10}} \left(\frac{\sqrt{7+2x} - (\sqrt{5} + \sqrt{2})}{x^2 - 10} \times \frac{\sqrt{7+2x} + (\sqrt{5} + \sqrt{2})}{\sqrt{7+2x} + (\sqrt{5} + \sqrt{2})} \right) \\
 &= \lim_{x \rightarrow \sqrt{10}} \left(\frac{7+2x - (\sqrt{5} + \sqrt{2})^2}{(x - \sqrt{10})(x + \sqrt{10})(\sqrt{7+2x} + (\sqrt{5} + \sqrt{2}))} \right) \\
 &= \lim_{x \rightarrow \sqrt{10}} \left(\frac{7+2x - (5+2+2\sqrt{10})}{(x - \sqrt{10})(x + \sqrt{10})(\sqrt{7+2x} + (\sqrt{5} + \sqrt{2}))} \right) \\
 &= \lim_{x \rightarrow \sqrt{10}} \left(\frac{2(x - \sqrt{10})}{(x - \sqrt{10})(x + \sqrt{10})(\sqrt{7+2x} + (\sqrt{5} + \sqrt{2}))} \right) \\
 &= \lim_{x \rightarrow \sqrt{10}} \left(\frac{2}{(x + \sqrt{10})(\sqrt{7+2x} + (\sqrt{5} + \sqrt{2}))} \right)
 \end{aligned}$$

$$= \left(\frac{2}{2\sqrt{10}(\sqrt{7} + 2\sqrt{10} + (\sqrt{5} + \sqrt{2}))} \right)$$

$$= \left(\frac{1}{(\sqrt{10}((\sqrt{5} + \sqrt{2})) + (\sqrt{5} + \sqrt{2}))} \right)$$

$$= \left(\frac{1}{2\sqrt{10}(\sqrt{5} + \sqrt{2})} \right)$$

$$= \left(\frac{1}{2(\sqrt{50} + \sqrt{20})} \right)$$

$$13. \lim_{x \rightarrow a} \left(\frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}} \right)$$

$$= \lim_{x \rightarrow a} \left(\frac{(a+2x-3x)(\sqrt{3a+x} + 2\sqrt{x})}{(3a+x-4x) \times (\sqrt{a+2x} + \sqrt{3x})} \right)$$

$$= \lim_{x \rightarrow a} \left(\frac{(a-x)(\sqrt{3a+x} + 2\sqrt{x})}{3(a-x) \times (\sqrt{a+2x} + \sqrt{3x})} \right)$$

$$= \lim_{x \rightarrow a} \left(\frac{(\sqrt{3a+x} + 2\sqrt{x})}{3(\sqrt{a+2x} + \sqrt{3x})} \right)$$

$$= \frac{1}{3} \left(\frac{2\sqrt{a} + 2\sqrt{a}}{\sqrt{3a} + \sqrt{3a}} \right)$$

$$= \frac{1}{3} \left(\frac{4}{2\sqrt{3}} \right) = \frac{2}{3\sqrt{3}}$$

$$14. \lim_{x \rightarrow -1} \frac{\sqrt[3]{7-x} - 2}{(x+1)}$$

$$= \lim_{x \rightarrow -1} \left(\frac{(7-x)^{1/3} - (8)^{1/3}}{(x+1)} \right)$$

$$= \lim_{x \rightarrow -1} \left(\frac{(7-x)^{1/3} - (8)^{1/3}}{(x+1)} \times \frac{(7-x)^{2/3} + 2(7-x)^{1/3} + 8^{2/3}}{(7-x)^{2/3} + 2(7-x)^{1/3} + 8^{2/3}} \right)$$

$$= \lim_{x \rightarrow -1} \left(\frac{(7-x-8)}{(x+1)} \times \frac{1}{(7-x)^{2/3} + 2(7-x)^{1/3} + 8^{2/3}} \right)$$

$$= \lim_{x \rightarrow -1} \left(\frac{-(1+x)}{(x+1)} \times \frac{1}{(7-x)^{2/3} + 2(7-x)^{1/3} + 8^{2/3}} \right)$$

$$= \lim_{x \rightarrow -1} \left(\frac{-1}{(7-x)^{2/3} + 2(7-x)^{1/3} + 8^{2/3}} \right)$$

$$= \left(\frac{-1}{(8)^{2/3} + 2(8)^{1/3} + 8^{2/3}} \right)$$

$$= \frac{-1}{4+4+4} = \frac{-1}{12}$$

$$15. \lim_{x \rightarrow 1} \left(\frac{\sqrt{x^2+8} - \sqrt{10-x^2}}{\sqrt{x^2+3} - \sqrt{5-x^2}} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{(x^2+8-10+x^2)(\sqrt{x^2+3} + \sqrt{5-x^2})}{(x^2+3-5+x^2)(\sqrt{x^2+8} + \sqrt{10-x^2})} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{(2x^2-2)(\sqrt{x^2+3} + \sqrt{5-x^2})}{(2x^2-2)(\sqrt{x^2+8} + \sqrt{10-x^2})} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{(\sqrt{x^2+3} + \sqrt{5-x^2})}{(\sqrt{x^2+8} + \sqrt{10-x^2})} \right)$$

$$= \frac{2+2}{3+3} = \frac{4}{6} = \frac{2}{3}$$

$$16. \lim_{x \rightarrow a} \left(\frac{x\sqrt{x} - a\sqrt{a}}{x-a} \right)$$

$$= \lim_{x \rightarrow a} \left(\frac{x^{2/3} - a^{2/3}}{x-a} \right)$$

$$= \frac{3}{2} (a)^{\frac{3}{2}-1}$$

$$= \frac{3\sqrt{a}}{2}$$

$$17. \lim_{x \rightarrow a} \left(\frac{(x+2)^{\frac{5}{3}} - (a+2)^{\frac{5}{3}}}{x-a} \right)$$

$$= \lim_{x+2 \rightarrow a+2} \left(\frac{(x+2)^{\frac{5}{3}} - (a+2)^{\frac{5}{3}}}{(x+2) - (a+2)} \right)$$

$$= \frac{5}{3} (a+2)^{\frac{5}{3}-1}$$

$$= \frac{5}{3} (a+2)^{\frac{2}{3}}$$

$$18. \lim_{x \rightarrow a} \left(\frac{x^m - a^m}{x^n - a^n} \right)$$

$$= \lim_{x \rightarrow a} \left(\frac{\frac{x^m - a^m}{x-a}}{\frac{x^n - a^n}{x-a}} \right)$$

$$= \frac{ma^{m-1}}{na^{n-1}} = \left(\frac{m}{n} \right) a^{m-n}$$

$$19. \lim_{x \rightarrow 0} \left(\frac{(1-x)^n - 1}{x} \right)$$

$$= \lim_{1-x \rightarrow 1} \left(\frac{(1-x)^n - 1}{(1-x) - 1} \right)$$

$$= n(1)^{n-1} = n$$

$$20. \lim_{x \rightarrow 1} \left(\frac{(x + x^2 + x^3 + \dots + x^n) - n}{x - 1} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{(x-1) + (x^2-1) + (x^3-1) + \dots + (x^n-1)}{x-1} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{(x-1)}{(x-1)} + \frac{(x^2-1)}{(x-1)} + \frac{(x^3-1)}{(x-1)} + \dots + \frac{(x^n-1)}{(x-1)} \right)$$

$$= (1 + 2 + 3 + 4 + \dots + n)$$

$$= \frac{n(n+1)}{2}$$

$$21. \lim_{x \rightarrow 1} \left(\frac{x^{n+1} - (n+1)x + n}{(x-1)^2} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{x^{n+1} - x - nx + n}{(x-1)^2} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{x(x^n - 1) - n(x-1)}{(x-1)^2} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{x(x-1)(1+x+x^2+x^3+\dots+x^{n-1}) - n(x-1)}{(x-1)^2} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{x(1+x+x^2+x^3+\dots+x^{n-1}) - n}{(x-1)} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{(x+x^2+x^3+\dots+x^n) - n}{(x-1)} \right)$$

$$= \frac{n(n+1)}{2}$$

$$22. \lim_{x \rightarrow 0} \left(\frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{(1+x^2)^{1/3} - (1-2x)^{1/4}}{x+x^2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\{(1+x^2)^{1/3} - 1\} - \{(1-2x)^{1/4} - 1\}}{x+x^2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\{(1+x^2)^{1/3} - 1\}}{x+x^2} \right) - \lim_{x \rightarrow 0} \left(\frac{\{(1-2x)^{1/4} - 1\}}{x+x^2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\{(1+x^2)^{1/3} - 1\}}{x^2} \times \frac{x^2}{x+x^2} \right)$$

$$- \lim_{x \rightarrow 0} \left(\frac{\{(1-2x)^{1/4} - 1\}}{-2x} \times \frac{-2x}{x+x^2} \right)$$

$$= \frac{1}{3} \times 0 + \frac{1}{4} \times 2 = \frac{1}{2}$$

$$23. \lim_{x \rightarrow 1} \left(\frac{\sqrt[3]{7+x^3} - \sqrt[2]{3+x^2}}{x-1} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{(7+x^3)^{1/3} - (3+x^2)^{1/2}}{x-1} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{\{(7+x^3)^{1/3} - 2\} - \{(3+x^2)^{1/2} - 2\}}{x-1} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{\{7+x^3\}^{1/3} - (8)^{1/3}}{x-1} \right)$$

$$- \lim_{x \rightarrow 1} \left(\frac{\{(3+x^2)^{1/2} - (4)^{1/2}\}}{x-1} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{\{(7+x^3-8)\}}{(x-1)\{(7+x)^{2/3} + 8^{2/3} + 8^{1/3}(7+x)^{1/3}\}} \right)$$

$$- \lim_{x \rightarrow 1} \left(\frac{\{(3+x^2-4)\}}{(x-1)(\sqrt{3+x^2} + 4)} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{(x^2+x+1)}{\{(7+x)^{2/3} + 8^{2/3} + 8^{1/3}(7+x)^{1/3}\}} \right)$$

$$- \lim_{x \rightarrow 1} \left(\frac{(x+1)}{(\sqrt{3+x^2} + 2)} \right)$$

$$= \frac{3}{3 \cdot 8^{2/3}} + \frac{1}{2}$$

$$= \frac{1}{4} + \frac{1}{2}$$

$$= \frac{3}{4}$$

$$24. \lim_{x \rightarrow 1} \left(\frac{\sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} - 3}{x-1} \right)$$

$$= \lim_{x \rightarrow 1} \left(\frac{(\sqrt[3]{x} - 1) + (\sqrt[4]{x} - 1) + (\sqrt[5]{x} - 1)}{x-1} \right)$$

$$= \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{20+15+12}{60} = \frac{47}{60}$$

$$\begin{aligned}
25. \quad & \lim_{x \rightarrow 1} \left(\frac{\left(\sum_{k=1}^{100} x^k \right) - 100}{x - 1} \right) \\
&= \lim_{x \rightarrow 1} \left(\frac{(x + x^2 + x^3 + \dots + x^{100}) - 100}{x - 1} \right) \\
&= \lim_{x \rightarrow 1} \left(\frac{(x - 1) + (x^2 - 1) + (x^3 - 1) + \dots + (x^{100} - 1)}{(x - 1)} \right) \\
&= \lim_{x \rightarrow 1} \left(\frac{(x - 1)}{(x - 1)} + \frac{(x^2 - 1)}{(x - 1)} + \frac{(x^3 - 1)}{(x - 1)} + \dots + \frac{(x^{100} - 1)}{(x - 1)} \right) \\
&= (1 + 2 + 3 + \dots + 100) \\
&= \frac{100 \times 101}{2} = 5050
\end{aligned}$$

$$\begin{aligned}
26. \quad & \lim_{n \rightarrow \infty} (4^n + 5^n)^{\frac{1}{n}} \\
&= \lim_{n \rightarrow \infty} \left(5^n \left(1 + \left(\frac{4}{5} \right)^n \right) \right)^{\frac{1}{n}} \\
&= 5.
\end{aligned}$$

$$\begin{aligned}
27. \quad & \lim_{n \rightarrow \infty} \left(\frac{n^2}{1 + 2 + 3 + \dots + n} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{2n^2}{n(n + 1)} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{2}{1 + 1/n} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{2}{1 + 1/n} \right) = 2
\end{aligned}$$

$$\begin{aligned}
28. \quad & \lim_{n \rightarrow \infty} \left(\frac{x^n + y^n}{x^n - y^n} \right), \text{ where } 0 < x < y. \\
&= \lim_{n \rightarrow \infty} \left(\frac{\left(\frac{x}{y} \right)^n + 1}{\left(\frac{x}{y} \right)^n - 1} \right) = \left(\frac{0 + 1}{0 - 1} \right) = -1
\end{aligned}$$

$$\begin{aligned}
29. \quad & \lim_{x \rightarrow \infty} (\sqrt{x^2 + 2x} - x) \\
&= \lim_{x \rightarrow \infty} \left(\frac{x^2 + 2x - x^2}{\sqrt{x^2 + 2x} + x} \right) \\
&= \lim_{x \rightarrow \infty} \left(\frac{2}{\sqrt{1 + \frac{2}{x}} + 1} \right) = \frac{2}{2} = 1
\end{aligned}$$

$$30. \quad \lim_{x \rightarrow \infty} (\sqrt{x^2 + 8x} + x)$$

$$= \lim_{y \rightarrow \infty} (\sqrt{y^2 - 8y} + y),$$

where $x = -y$, $x \rightarrow -\infty$, $y \rightarrow \infty$

$$= \lim_{y \rightarrow \infty} \left(\frac{y^2 - 8y - y^2}{\sqrt{y^2 - 8y} + y} \right)$$

$$= \lim_{y \rightarrow \infty} \left(\frac{-8y}{\sqrt{y^2 - 8y} + y} \right)$$

$$= \lim_{y \rightarrow \infty} \left(\frac{-8}{\sqrt{1 - \frac{8}{y}} + 1} \right) = -4$$

$$31. \quad \lim_{x \rightarrow \infty} \left(\frac{x^2 + 2x - 1}{2x^2 - 3x - 2} \right)^{\frac{2x+5}{2x-7}}$$

$$= \frac{1}{2}, \text{ where } \lim_{x \rightarrow \infty} \left(\frac{2x+5}{2x-7} \right) = 1$$

$$\text{and } \lim_{x \rightarrow \infty} \left(\frac{x^2 + 2x - 1}{2x^2 - 3x - 2} \right) = \frac{1}{2}$$

$$\begin{aligned}
32. \quad & \lim_{x \rightarrow \infty} \left(\frac{(x+1)^{10} + (x+2)^{10} + \dots + (x+2012)^{10}}{x^{10} + 2012^{10}} \right) \\
&= \lim_{x \rightarrow \infty} \left(\frac{(1 + 1/x)^{10} + (1 + 2/x)^{10} + \dots + (1 + 2012/x)^{10}}{1 + (2012/x)^{10}} \right) \\
&= 2012
\end{aligned}$$

$$\begin{aligned}
33. \quad & \text{Let } t_r = (n - r + 1) \cdot r \\
&= (nr - (r - 1)r) \\
&= ((n - 1)(r - r^2))
\end{aligned}$$

$$\begin{aligned}
\text{Now, } S_n &= \sum_{r=1}^n t_r \\
&= \sum_{r=1}^n ((n - 1)r - r^2) \\
&= \sum_{r=1}^n (n - 1)r - \sum_{r=1}^n r^2 \\
&= (n - 1) \left(\frac{n(n + 1)}{2} \right) - \frac{n(n + 1)(2n + 1)}{6} \\
&= \left(\frac{n(n + 1)}{2} \right) \left(n - 1 - \frac{2n + 1}{3} \right) \\
&= \left(\frac{n(n + 1)(n - 2)}{6} \right)
\end{aligned}$$

$$\text{Therefore, } \lim_{n \rightarrow \infty} \left(\frac{n(n + 1)(n - 2)}{6n^3} \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{\left(1 + \frac{1}{n}\right) \left(1 - \frac{2}{n}\right)}{6} \right)$$

$$= \frac{1}{6}$$

34. Put $\frac{1}{n} = x$

$$\lim_{x \rightarrow 0} \left\{ \frac{\sqrt{1 - \cos x} \sqrt{1 - \cos x} \sqrt{1 - \cos x} \dots \infty}{x^2} \right\}$$

$$= \lim_{x \rightarrow 0} \left(\frac{(1 - \cos x)^{\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots \infty}}{x^2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{(1 - \cos x)}{x^2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{2 \sin^2(x/2)}{x^2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{2 \sin^2(x/2)}{(x^2/4)/4} \right)$$

$$= \frac{1}{2}$$

35. Given $\lim_{x \rightarrow \infty} \left(\frac{x^2 - 1}{x + 1} - ax - b \right) = 2$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{x^2 - 1 - ax^2 - ax - bx - b}{x + 1} \right) = 2$$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{(1 - a)x^2 - (a + b)x - (1 + b)}{x + 1} \right) = 2$$

It is possible only when $1 - a = 0 \Rightarrow a = 1$

Also, $\lim_{x \rightarrow \infty} \left(\frac{-(a + b)x - (1 + b)}{x + 1} \right) = 2$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{-(a + b) - \frac{(1 + b)}{x}}{1 + \frac{1}{x}} \right) = 2$$

$$\Rightarrow -(a + b) = 2$$

$$\Rightarrow (a + b) = -2$$

$$\Rightarrow b = -a - 2 = -1 - 2 = -3$$

36. Given $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 1}{x + 1} - ax - b \right) = 0$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{x^2 + 1 - ax^2 - ax - bx - b}{x + 1} \right) = 0$$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{(1 - a)x^2 - (a + b)x + (1 - b)}{x + 1} \right) = 0$$

It is possible only when, $1 - a = 0$ and $(a + b) = 0$

Thus, $a = 1$ and $b = -1$

37. Given $\lim_{x \rightarrow 0} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$

$$\Rightarrow \lim_{x \rightarrow 0} \left(\frac{x^2 + x + 1 - ax^2 - ax - bx - b}{x + 1} \right) = 4$$

$$\Rightarrow \lim_{x \rightarrow 0} \left(\frac{(1 - a)x^2 + (1 - a - b)x + (1 - b)}{x + 1} \right) = 4$$

It is possible only when

$$\Rightarrow (1 - a) = 0, (1 - a - b) = 4$$

$$\Rightarrow a = 1, a + b = -3$$

$$\Rightarrow a = 1 \text{ and } b = -4$$

38. We have

$$\lim_{x \rightarrow -\infty} \left(\frac{x^4 \sin\left(\frac{1}{x}\right) + x^2}{1 + |x|^3} \right)$$

$$= \lim_{x \rightarrow -\infty} \left(\frac{x^4 \sin\left(\frac{1}{x}\right) + x^2}{1 - x^3} \right)$$

$$= \lim_{x \rightarrow -\infty} \left(\frac{x \sin\left(\frac{1}{x}\right) + \frac{1}{x}}{\frac{1}{x^3} - 1} \right)$$

$$= \left(\frac{1 + 0}{0 - 1} \right)$$

$$= -1$$

39. Limit = $\lim_{x \rightarrow 0} \left(\frac{\sin 3x}{5x} \right)$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{3x} \times \frac{3x}{5x} \right)$$

$$= \frac{3}{5}$$

40. Limit = $\lim_{x \rightarrow 0} \left(\frac{\sin ax}{\sin bx} \right)$

$$= \lim_{x \rightarrow 0} \left(\frac{\frac{\sin ax}{ax} \times ax}{\frac{\sin bx}{bx} \times bx} \right)$$

$$= \frac{a}{b}$$

$$\begin{aligned}
 41. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{\sin 2x + 3x}{4x + \sin 6x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\frac{\sin 2x}{x} + 3}{4 + \frac{\sin 6x}{x}} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\frac{\sin 2x}{2x} \times 2 + 3}{4 + \frac{\sin 6x}{6x} \times 6} \right) \\
 &= \left(\frac{2 + 3}{4 + 6} \right) \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 42. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{\sin 2x + \sin 5x}{\sin 4x + \sin 6x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\frac{\sin 2x}{x} + \frac{\sin 5x}{x}}{\frac{\sin 4x}{x} + \frac{\sin 6x}{x}} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\frac{\sin 2x}{2x} \times 2 + \frac{\sin 5x}{5x} \times 5}{\frac{\sin 4x}{4x} \times 4 + \frac{\sin 6x}{6x} \times 6} \right) \\
 &= \left(\frac{2 + 5}{4 + 6} \right) \\
 &= \frac{7}{10}
 \end{aligned}$$

$$\begin{aligned}
 43. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{1 - \cos 4x}{1 - \cos 6x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{2 \sin^2 2x}{2 \sin^2 3x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\frac{\sin^2 2x}{4x^2} \times 4x^2}{\frac{\sin^2 3x}{9x^2} \times 9x^2} \right) \\
 &= \frac{4}{9}
 \end{aligned}$$

$$\begin{aligned}
 44. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{\sqrt{1 - \cos 2x}}{x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\sqrt{2 \sin^2 x}}{x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\sqrt{2} |\sin x|}{x} \right)
 \end{aligned}$$

$$\begin{aligned}
 &= \begin{cases} \lim_{x \rightarrow 0^+} \left(\frac{\sqrt{2} |\sin x|}{x} \right) \\ \lim_{x \rightarrow 0^-} \left(\frac{\sqrt{2} |\sin x|}{x} \right) \end{cases} \\
 &= \begin{cases} \lim_{x \rightarrow 0^+} \left(\frac{\sqrt{2} (\sin x)}{x} \right) \\ \lim_{x \rightarrow 0^-} \left(\frac{\sqrt{2} (\sin x)}{x} \right) \end{cases} \\
 &= \begin{cases} \sqrt{2} \\ -\sqrt{2} \end{cases}
 \end{aligned}$$

Since R.H.L $\neq$ L.H.L, so limit does not exist.

$$\begin{aligned}
 45. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{\cos 7x - \cos 9x}{\cos 3x - \cos 5x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{2 \sin 8x \sin x}{2 \sin 4x \sin x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\sin 8x}{\sin 4x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\frac{\sin 8x}{8x} \times 8x}{\frac{\sin 4x}{4x} \times 4x} \right) \\
 &= 2
 \end{aligned}$$

$$\begin{aligned}
 46. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{1 - \cos x \sqrt{\cos 2x}}{x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{(1 - \cos x \sqrt{\cos 2x})(1 + \cos x \sqrt{\cos 2x})}{x^2(1 + \cos x \sqrt{\cos 2x})} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{(1 - \cos^2 x (\cos 2x))}{x^2(1 + \cos x \sqrt{\cos 2x})} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{(1 - \cos^2 x (2 \cos^2 x - 1))}{x^2(1 + \cos x \sqrt{\cos 2x})} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{(1 - 2 \cos^4 x + \cos^2 x)}{x^2(1 + \cos x \sqrt{\cos 2x})} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{(2 \cos^4 x - \cos^2 x - 1)}{x^2(1 + \cos x \sqrt{\cos 2x})} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\sin^2 x (2 \cos^2 x + 1)}{x^2(1 + \cos x \sqrt{\cos 2x})} \right)
 \end{aligned}$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin^2 x}{x^2} \times \frac{(2 \cos^2 x + 1)}{(1 + \cos x \sqrt{\cos 2x})} \right)$$

$$= \frac{2}{3}$$

$$47. \text{ Limit} = \lim_{x \rightarrow 0} \left(\frac{\tan x - \sin x}{x^3} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin x (1 - \cos x)}{x^3 \cos x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \times \frac{2 \sin^2 \left(\frac{x}{2} \right)}{\frac{x^2}{4} \times 4 \cos x} \right)$$

$$= \frac{1}{2}$$

$$48. \text{ We have } \cos x \cos 2x \cos 3x$$

$$= \frac{1}{2} (2 \cos 3x \cos x) \cos 2x$$

$$= \frac{1}{2} (\cos 4x + \cos 2x) \cos 2x$$

$$= \frac{1}{4} (2 \cos 4x \cos 2x + 2 \cos^2 2x)$$

$$= \frac{1}{4} (\cos 6x + \cos 2x + 1 + \cos 4x)$$

$$= \frac{1}{4} (1 + \cos 2x + \cos 4x + \cos 6x)$$

Thus

$$\text{Limit} = \lim_{x \rightarrow 0} \left(\frac{1 - \cos x \cos 2x \cos 3x}{\sin^2 2x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{1 - \frac{1}{4}(1 + \cos 2x + \cos 4x + \cos 6x)}{\sin^2 2x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{3 - (\cos 2x + \cos 4x + \cos 6x)}{4 \sin^2 2x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{(1 - \cos 6x) + (1 - \cos 4x) + (1 - \cos 2x)}{4 \sin^2 2x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{2 \sin^2 3x + 2 \sin^2 2x + 2 \sin^2 x}{4 \sin^2 2x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin^2 3x + \sin^2 2x + \sin^2 x}{2 \sin^2 2x} \right)$$

$$= \frac{9 + 4 + 1}{2 \times 4} = \frac{14}{8} = \frac{7}{4}$$

$$49. \text{ Limit} = \lim_{x \rightarrow 0} \frac{\sin^{-1} x - \tan^{-1} x}{x^3}$$

$$= \lim_{x \rightarrow 0} \left(\frac{\tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right) - \tan^{-1} x}{x^3} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\tan^{-1} \left(\frac{\frac{x}{\sqrt{1-x^2}} - x}{1 + x \cdot \frac{x}{\sqrt{1-x^2}}} \right)}{x^3} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\tan^{-1} \left(\frac{x - x\sqrt{1-x^2}}{\sqrt{1-x^2} + x^2} \right)}{x^3} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\tan^{-1} \left(\frac{x - x\sqrt{1-x^2}}{\sqrt{1-x^2} + x^2} \right)}{\left(\frac{x - x\sqrt{1-x^2}}{\sqrt{1-x^2} + x^2} \right)^3} \times \left(\frac{x - x\sqrt{1-x^2}}{\sqrt{1-x^2} + x^2} \right) \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{x(1 - \sqrt{1-x^2})}{x^3(\sqrt{1-x^2} + x^2)} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{(1 - 1 + x^2)(\sqrt{1-x^2} - x^2)}{x^2(1 + \sqrt{1-x^2})(1 - x^2 - x^2)} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{(x^2)(\sqrt{1-x^2} - x^2)}{x^2(1 + \sqrt{1-x^2})(1 - 2x^2)} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{(\sqrt{1-x^2} - x^2)}{(1 + \sqrt{1-x^2})(1 - 2x^2)} \right)$$

$$= \frac{1}{2}$$

$$50. \text{ Limit} = \lim_{x \rightarrow 0} \left(\frac{x - \sin x}{x^3} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{x - \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right)}{x^3} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\left(\frac{x^3}{3!} - \frac{x^5}{5!} + \frac{x^7}{7!} - \dots \right)}{x^3} \right)$$

$$= \frac{1}{6}$$

$$\begin{aligned}
 51. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{x - \tan x}{x^3} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{x - \left(x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots \right)}{x^3} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{-\left(\frac{x^3}{3} + \frac{2}{15}x^5 + \dots \right)}{x^3} \right) \\
 &= -\frac{1}{3}
 \end{aligned}$$

$$\begin{aligned}
 52. \text{ Limit} &= \lim_{x \rightarrow \infty} \left(\frac{x + \sin x}{x + \cos x} \right) \\
 &= \lim_{x \rightarrow \infty} \left(\frac{1 + \frac{\sin x}{x}}{1 + \frac{\cos x}{x}} \right) \\
 &= \left(\frac{1 + 0}{1 + 0} \right) \\
 &= 1
 \end{aligned}$$

53. As we know that, $\sin x \leq x$ and $\tan x \geq x$

$$\text{Thus, } \frac{\sin x}{x} \leq 1 \text{ and } \frac{\tan x}{x} \geq 1$$

Therefore,

$$\begin{aligned}
 \text{Limit} &= \lim_{x \rightarrow 0} \left(\left[\frac{n \sin x}{x} \right] + \left[\frac{n \tan x}{x} \right] \right) \\
 &= (n - 1) + n \\
 &= 2n - 1.
 \end{aligned}$$

$$\begin{aligned}
 54. \text{ We have } a &= \min \{x^2 + 2x + 3, x^2 + 4x + 10\} \\
 a &= \min \{(x + 1)^2 + 2, (x + 2)^2 + 6\} \\
 a &= 2
 \end{aligned}$$

$$\text{Also, } b = \lim_{x \rightarrow 0} \left(\frac{1 - \cos \theta}{\theta^2} \right)$$

$$\begin{aligned}
 &= \lim_{\theta \rightarrow 0} \left(\frac{2 \sin^2 \left(\frac{\theta}{2} \right)}{\left(\frac{\theta^2}{4} \right) \times 4} \right) \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{Therefore, } \sum_{r=0}^n (a^r b^{n-r}) &= \sum_{r=0}^n \left(2^r \left(\frac{1}{2} \right)^{n-r} \right) \\
 &= \frac{1}{2^n} \sum_{r=0}^n (4^r) \\
 &= \frac{1}{2^n} (1 + 4 + 4^2 + \dots + 4^n)
 \end{aligned}$$

$$= \frac{1}{2^n} \left(\frac{4^{n+1} - 1}{3} \right)$$

$$= \left(\frac{4^{n+1} - 1}{3 \cdot 2^n} \right)$$

$$55. \text{ Given } \lim_{x \rightarrow 0} \left(\frac{a \cos x + bx \sin x - 5}{x^4} \right)$$

As $x \rightarrow 0$, $x^4 \rightarrow 0$, the limit must be in the form of $\left(\frac{0}{0} \right)$

$$\text{Thus, } \lim_{x \rightarrow 0} (a \cos x + bx \sin x - 5) = 0$$

$$\Rightarrow a = 5 = 0$$

$$\Rightarrow a = 5$$

Now, Limit

$$= \lim_{x \rightarrow 0} \left(\frac{a \cos x + bx \sin x - 5}{x^4} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{5 \cos x + bx \sin x - 5}{x^4} \right)$$

$$= \lim_{x \rightarrow 0} \frac{5 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) + bx \left(x - \frac{x^3}{3!} + \dots \right) - 5}{x^4}$$

$$= \lim_{x \rightarrow 0} \frac{\left(b - \frac{5}{2} \right) x^2 + \left(\frac{5}{24} - \frac{b}{6} \right) x^4 + \dots}{x^4}$$

It will provide finite limit only when, co-efficient of x^2 is zero.

$$\text{Thus, } \left(b - \frac{5}{2} \right) = 0$$

$$\Rightarrow b = \frac{5}{2}$$

Also, the limit value

$$= \lim_{x \rightarrow 0} \frac{\left(\frac{5}{24} - \frac{b}{6} \right) x^4 + \dots}{x^4}$$

$$= \left(\frac{5}{24} - \frac{b}{6} \right)$$

$$= \left(\frac{5}{24} - \frac{5}{12} \right)$$

$$= -\frac{5}{24}$$

Therefore, the value of $a + 2b + 10$

$$= 5 + 5 + 10$$

$$= 20.$$

$$\begin{aligned}
 56. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{\sin(\pi \cos^2 x)}{x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\sin(\pi(1 - \sin^2 x))}{x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\sin((\pi - \pi \sin^2 x))}{x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\sin(\pi \sin^2 x)}{x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\sin(\pi \sin^2 x)}{\pi \sin^2 x} \times \frac{\pi \sin^2 x}{x^2} \right) \\
 &= \pi
 \end{aligned}$$

$$\begin{aligned}
 57. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{e^{4x} - 1}{5x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^{4x} - 1}{5x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^{4x} - 1}{4x} \times \frac{4x}{5x} \right) \\
 &= \frac{4}{5}
 \end{aligned}$$

$$\begin{aligned}
 58. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{e^{3x} - 1}{e^{5x} - 1} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\frac{e^{3x} - 1}{3x} \times 3x}{\frac{e^{5x} - 1}{5x} \times 5x} \right) \\
 &= \frac{3}{5}
 \end{aligned}$$

$$\begin{aligned}
 59. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{e^x + e^{-x} - 2}{x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^x + \frac{1}{e^x} - 2}{x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{(e^x - 1)^2}{e^x x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\left(\frac{e^x - 1}{x} \right)^2 \times \frac{1}{e^x} \right) \\
 &= 1.
 \end{aligned}$$

$$60. \text{ Limit} = \lim_{x \rightarrow 0} \left(\frac{e^{3+x} - e^3}{x} - \frac{\sin x}{x} \right)$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \left(\frac{e^3(e^x - 1)}{x} - \frac{\sin x}{x} \right) \\
 &= (e^3 - 1)
 \end{aligned}$$

$$\begin{aligned}
 61. \text{ Limit} &= \lim_{x \rightarrow 0} \frac{e^x - e^{\sin x}}{x - \sin x} \\
 &= \lim_{x \rightarrow 0} e^{\sin x} \left(\frac{e^{x - \sin x} - 1}{x - \sin x} \right) \\
 &= e^0 \times 1 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 62. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{e^x - e^{x \cos x}}{x + \sin x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^{x \cos x} (e^{x - x \cos x} - 1)}{x + \sin x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^{x \cos x} (e^{x - x \cos x} - 1)}{(x - x \cos x)} \times \frac{x(1 - \cos x)}{x + \sin x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^{x \cos x} (e^{x - x \cos x} - 1)}{(x - x \cos x)} \times \frac{(1 - \cos x)}{1 + \frac{\sin x}{x}} \right) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 63. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{e^x - 1 - x}{x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right) - 1 - x}{x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\left(\frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right)}{x^2} \right) \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 64. \text{ Limit} &= \lim_{x \rightarrow 0} \frac{8^x - 4^x - 2^x + 1}{x^2} \\
 &= \lim_{x \rightarrow 0} \left(\frac{4^x(2^x - 1) - (2^x - 1)}{x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{(4^x - 1)(2^x - 1)}{x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\left(\frac{4^x - 1}{x} \right) \left(\frac{2^x - 1}{x} \right) \right) \\
 &= \log 4 \times \log 2 \\
 &= 2(\log 2)^2
 \end{aligned}$$

$$\begin{aligned}
65. \text{ Limit} &= \lim_{x \rightarrow 0} \frac{9^x - 2 \cdot 6^x + 4^x}{x^2} \\
&= \lim_{x \rightarrow 0} \left(\frac{(3^x)^2 - 2 \cdot 3^x \cdot 2^x + (2^x)^2}{x^2} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{(3^x - 2^x)^2}{x^2} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{3^x - 2^x}{x} \right)^2 \\
&= \lim_{x \rightarrow 0} \left(\frac{3^x - 1}{x} - \frac{2^x - 1}{x} \right)^2 \\
&= (\log 3 - \log 2)^2 \\
&= \log^2 \left(\frac{3}{2} \right)
\end{aligned}$$

$$\begin{aligned}
66. \text{ Limit} &= \lim_{x \rightarrow a} \left(\frac{a^x - a^a}{x - a} \right), a > 0 \\
&= \lim_{x \rightarrow a} \left(\frac{a^a (a^{x-a} - 1)}{x - a} \right) \\
&= (a^a \cdot \log a)
\end{aligned}$$

$$\begin{aligned}
67. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{a^{x+h} + a^{x-h} - 2a^x}{h^2} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{a^x (a^h + a^{-h} - 2)}{h^2} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{a^x (a^{2h} - 2a^h + 1)}{a^h \times h^2} \right) \\
&= \lim_{h \rightarrow 0} \left(\frac{a^x}{a^h} \times \left(\frac{a^{2h} - 2a^h + 1}{h^2} \right) \right) \\
&= \lim_{h \rightarrow 0} \left(\frac{a^x}{a^h} \times \left(\frac{a^h - 1}{h} \right)^2 \right) \\
&= a^x \log a
\end{aligned}$$

$$\begin{aligned}
68. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{8^x - 7^x}{6^x - 5^x} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{(8^x - 1) - (7^x - 1)}{(6^x - 1) - (5^x - 1)} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{\left(\frac{8^x - 1}{x} \right) - \left(\frac{7^x - 1}{x} \right)}{\left(\frac{6^x - 1}{x} \right) - \left(\frac{5^x - 1}{x} \right)} \right) \\
&= \frac{(\log 8 - \log 7)}{(\log 6 - \log 5)} \\
&= \left(\frac{\log(8/7)}{\log(6/5)} \right)
\end{aligned}$$

$$\begin{aligned}
69. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{(5^x - 1)(4^x - 1)}{(3^x - 1)(6^x - 1)} \right) \\
&= \lim_{x \rightarrow 0} \frac{\left(\frac{5^x - 1}{x} \right) \left(\frac{4^x - 1}{x} \right)}{\left(\frac{3^x - 1}{x} \right) \left(\frac{6^x - 1}{x} \right)} \\
&= \frac{(\log 5 \cdot \log 4)}{(\log 3 \cdot \log 6)}
\end{aligned}$$

$$\begin{aligned}
70. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{x \cdot 2^x - x}{1 - \cos x} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{x(2^x - 1)}{1 - \cos x} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{x^2 \left(\frac{2^x - 1}{x} \right)}{\frac{2 \sin^2 \left(\frac{x}{2} \right)}{\frac{x^2}{4}} \times \frac{x^2}{4}} \right) \\
&= 2 \log 2
\end{aligned}$$

$$\begin{aligned}
71. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{e^x - e^{-x} - 2x}{x - \sin x} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{\left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right) - \left(1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \right) - 2x}{x - \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right)} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{\left(\frac{x^3}{3!} + \frac{x^4}{4!} + \dots \right) - \left(-\frac{x^3}{3!} + \frac{x^4}{4!} - \dots \right)}{-\left(-\frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right)} \right) \\
&= \left(\frac{\frac{1}{3!} + \frac{1}{3!}}{\frac{1}{3!}} \right) \\
&= 2
\end{aligned}$$

$$\begin{aligned}
72. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{e^{x^3} - 1 - x^3}{\sin^6 2x} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{e^{x^3} - 1 - x^3}{\left(\frac{\sin^6 2x}{(2x)^6} \times (2x)^6 \right)} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{e^{x^3} - 1 - x^3}{(2x)^6} \right) \left(\because \frac{\sin^6 2x}{(2x)^6} = 1 \right)
\end{aligned}$$

$$= \lim_{t \rightarrow 0} \left(\frac{e^t - 1 - t}{2^6 \cdot t^2} \right)$$

(put $x^3 = t$, when $x \rightarrow 0$, $t \rightarrow 0$)

$$= \lim_{t \rightarrow 0} \left(\frac{1}{2^6} \times \left(\frac{e^t - 1 - t}{t^2} \right) \right)$$

$$= \frac{1}{2^6} \times \frac{1}{2}$$

$$= \left(\frac{1}{128} \right)$$

$$73. \text{ Limit} = \lim_{x \rightarrow 2} \left(\frac{(\cos \alpha)^x + (\sin \alpha)^x - 1}{x - 2} \right)$$

$$= \lim_{x \rightarrow 2} \left(\frac{(\cos \alpha)^x + (\sin \alpha)^x - (\cos^2 \alpha + \sin^2 \alpha)}{x - 2} \right)$$

$$= \lim_{x \rightarrow 2} \left(\frac{((\cos \alpha)^x - \cos^2 \alpha) + (\sin \alpha)^x - \sin^2 \alpha}{x - 2} \right)$$

$$= \lim_{t \rightarrow 0} \left(\frac{((\cos \alpha)^{t+2} - \cos^2 \alpha) + (\sin \alpha)^{t+2} - \sin^2 \alpha}{t} \right)$$

$$= \lim_{t \rightarrow 0} \left(\cos^2 \alpha \left(\frac{(\cos \alpha)^t - 1}{t} \right) + \sin^2 \alpha \left(\frac{(\sin \alpha)^t - 1}{t} \right) \right)$$

$$= (\cos^2 \alpha \log(\cos \alpha) + \sin^2 \alpha \log(\sin \alpha))$$

$$74. \text{ Limit} = \lim_{x \rightarrow 0} \left(\frac{e^{x \cos x} - 1 - x}{\sin(x^2)} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{e^{x \cos x} - 1}{\sin(x^2)} - \frac{x}{(\sin(x^2))} \right)$$

$$75. \text{ We have } a = \min\{x^2 + 4x + 6, x^2 + 2x + 8\}$$

$$a = \min\{(x+2)^2 + 2, (x+1)^2 + 5\}$$

$$a = 2$$

$$\text{Also, } b = \lim_{x \rightarrow 0} \left(\frac{\sin x \cos x}{e^x - e^{-x}} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\frac{\sin x}{x} \times \cos x}{\left(\frac{e^x - 1}{x} \right) - \left(\frac{e^{-x} - 1}{-x} \right)} \right) = \frac{1}{2}$$

$$\text{Thus, } \sum_{r=0}^n (a^r b^{n-r})$$

$$= \sum_{r=0}^n \left(2^r \left(\frac{1}{2} \right)^{n-r} \right)$$

$$= \frac{1}{2^n} \sum_{r=0}^n (4^r)$$

$$= \frac{1}{2^n} (1 + 4 + 4^2 + \dots + 4^n)$$

$$= \frac{1}{2^n} \left(\frac{4^{n+1} - 1}{3} \right)$$

$$= \left(\frac{4^{n+1} - 1}{3 \cdot 2^n} \right)$$

$$76. \text{ We have } \lim_{x \rightarrow 0} \left(\frac{Ae^x - B \cos x + Ce^{-x}}{x \sin x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{A \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right) - B \left(1 - \frac{x^2}{2!} + \dots \right) + C \left(1 - x + \frac{x^2}{2!} + \dots \right)}{x \left(x - \frac{x^3}{3!} + \dots \right)} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{(A - B + C) + (A - C)x + \left(\frac{A + B + C}{2} \right)x^2 + \dots}{x \left(x - \frac{x^3}{3!} + \dots \right)} \right)$$

It will provide finite limit only when, $A - B + C = 0$,

$$A - C = 0 \text{ and } \frac{A + B + C}{2} = 2$$

On solving, we get, $A = 1$, $B = 2$, $C = 1$

Hence, the value of $A + B + C + 10 = 14$.

$$77. \text{ Limit} = \lim_{x \rightarrow 0} \left(\frac{\log(1 + 3x)}{\sin 2x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\frac{\log(1 + 3x)}{3x} \times 3x}{\frac{\sin 2x}{2x} \times 2x} \right)$$

$$= \frac{3}{2}$$

$$78. \text{ Limit} = \lim_{x \rightarrow 0} \left(\frac{\log(1 + 3x)}{\log(1 - 2x)} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\frac{\log(1 + 3x)}{3x} \times 3x}{\frac{\log(1 - 2x)}{-2x} \times -2x} \right)$$

$$= -\frac{3}{2}$$

$$79. \text{ Limit} = \lim_{x \rightarrow e} \left(\frac{\log x - 1}{x - e} \right)$$

$$= \lim_{x \rightarrow e} \left(\frac{\log x - \log e}{x - e} \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{\log(y+e) - \log e}{y+e-e} \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{\log\left(\frac{y+e}{e}\right)}{y} \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{\log\left(1 + \frac{y}{e}\right)}{\frac{y}{e} \times e} \right)$$

$$= \frac{1}{e}$$

$$80. \text{ Limit} = \lim_{x \rightarrow 5} \left(\frac{\log x - \log 5}{x - 5} \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{\log(y+5) - \log 5}{y+5-5} \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{\log\left(\frac{y+5}{5}\right)}{y} \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{\log\left(1 + \frac{y}{5}\right)}{\frac{y}{5} \times 5} \right)$$

$$= \frac{1}{5}$$

$$81. \text{ Limit} = \lim_{x \rightarrow 5} \left(\frac{\log(x+5) - \log(5-x)}{x-5} \right)$$

$$= \lim_{x \rightarrow 5} \left(\frac{(\log(x+5) - \log 5) - (\log(5-x) - \log 5)}{x-5} \right)$$

$$= \lim_{x \rightarrow 5} \left(\left(\frac{\log(x+5) - \log 5}{x-5} \right) - \left(\frac{\log(5-x) - \log 5}{x-5} \right) \right)$$

$$= \lim_{x \rightarrow 5} \left(\left(\frac{\log(x+5) - \log 5}{x-5} \right) + \left(\frac{\log(5-x) - \log 5}{5-x} \right) \right)$$

$$= \frac{1}{5} + \frac{1}{5} = \frac{2}{5}$$

$$82. \text{ Limit} = \lim_{x \rightarrow 0} \left(\frac{e^x - \log(x+e)}{e^x - 1} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{e^x - \log\left\{e\left(1 + \frac{x}{e}\right)\right\}}{e^x - 1} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{e^x - 1 - \left\{\log\left(1 + \frac{x}{e}\right)\right\}}{e^x - 1} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\frac{e^x - 1}{x} - \left\{ \frac{\log\left(1 + \frac{x}{e}\right)}{\frac{x}{e} \times e} \right\}}{\frac{e^x - 1}{x}} \right)$$

$$= \left(1 - \frac{1}{e}\right)$$

$$83. \text{ Limit} = \lim_{x \rightarrow 0} \left(\frac{\ln(\cos x)}{\sqrt[4]{1+x^2} - 1} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\log\left(1 - 2\sin^2\left(\frac{x}{2}\right)\right)}{-2\sin^2\left(\frac{x}{2}\right)} \times \frac{-2\sin^2\left(\frac{x}{2}\right)}{\sqrt[4]{x^2+1} - 1} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{-2\sin^2\left(\frac{x}{2}\right)}{\sqrt[4]{x^2+1} - 1} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{-2\sin^2\left(\frac{x}{2}\right) \times (\sqrt[4]{x^2+1} + 1)}{(\sqrt[4]{x^2+1} - 1)(\sqrt[4]{x^2+1} + 1)} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{-2\sin^2\left(\frac{x}{2}\right) \times (\sqrt[4]{x^2+1} + 1)(\sqrt[4]{x^2+1} + 1)}{(\sqrt[4]{x^2+1} - 1)(\sqrt[4]{x^2+1} + 1)} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{-2\sin^2\left(\frac{x}{2}\right) \times (\sqrt[4]{x^2+1} + 1)(\sqrt[4]{x^2+1} + 1)}{(x^2)} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{-2\sin^2\left(\frac{x}{2}\right) \times (\sqrt[4]{x^2+1} + 1)(\sqrt[4]{x^2+1} + 1)}{\left(\frac{x^2}{4}\right) \times 4} \right)$$

$$= -2$$

$$84. \text{ Limit} = \lim_{x \rightarrow 0} \left(\frac{\ln(\cos(\sin x))}{x^2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\log\left(1 - 2\sin^2\left(\frac{\sin x}{2}\right)\right)}{-2\sin^2\left(\frac{\sin x}{2}\right)} \times \frac{-2\sin^2\left(\frac{\sin x}{2}\right)}{x^2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{-2\sin^2\left(\frac{\sin x}{2}\right)}{x^2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{-2 \sin^2 \left(\frac{\sin x}{2} \right)}{\left(\frac{\sin^2 x}{4} \right)} \times \frac{\sin^2 x}{x^2} \times \frac{1}{4} \right)$$

$$= -\frac{1}{2}$$

85. Limit = $\lim_{x \rightarrow \infty} \left(x - x^2 \ln \left(1 + \frac{1}{x} \right) \right)$

$$= \lim_{y \rightarrow 0} \left(\frac{1}{y} - \frac{1}{y^2} \log(1 + y) \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{1}{y} - \frac{1}{y^2} \left(y - \frac{y^2}{2} + \frac{y^3}{3} - \frac{y^4}{4} + \dots \right) \right)$$

$$= \lim_{y \rightarrow 0} \left(- \left(-\frac{1}{2} + \frac{y}{3} - \frac{y^2}{4} + \dots \right) \right)$$

$$= \frac{1}{2}$$

86. Limit = $\lim_{x \rightarrow 1} \left(\frac{x^x - 1}{x \ln x} \right)$

$$= \lim_{y \rightarrow 0} \left(\frac{(1 + y)^{1+y} - 1}{(1 + y) \log(1 + y)} \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{1 + y(1 + y) + y(1 + y) \frac{y^2}{2!} + \dots - 1}{(1 + y) \frac{\log(1 + y)}{y} \times y} \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{y(1 + y) + y(1 + y) \frac{y^2}{2!} + \dots}{(1 + y)y} \right)$$

$$= \lim_{y \rightarrow 0} \left(1 + \frac{y^2}{2!} + (y - 1) \frac{y^3}{3!} + \dots \right)$$

$$= 1.$$

87. Limit = $\lim_{x \rightarrow 1} \left(\frac{1}{\ln x} - \frac{1}{x - 1} \right)$

$$= \lim_{y \rightarrow 0} \left(\frac{1}{\log(1 + y)} - \frac{1}{y} \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{y - \log(1 + y)}{y \log(1 + y)} \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{y - \log(1 + y)}{y^2 \times \frac{\log(1 + y)}{y}} \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{y - \log(1 + y)}{y^2} \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{y - \left(y - \frac{y^2}{2} + \frac{y^3}{3} - \frac{y^4}{4} + \dots \right)}{y^2} \right)$$

$$= \lim_{y \rightarrow 0} \left(- \left(-\frac{1}{2} + \frac{y}{3} - \frac{y^2}{4} + \dots \right) \right)$$

$$= \frac{1}{2}$$

88. Limit = $\lim_{x \rightarrow \pi/4} \left(\frac{\ln \cot x}{1 - \cot x} \right)$

$$= \lim_{y \rightarrow 0} \left(\frac{\log \left(\cot \left(\frac{\pi}{4} + y \right) \right)}{1 - \cot \left(\frac{\pi}{4} + y \right)} \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{\log \left(\frac{\cot y - 1}{\cot y + 1} \right)}{1 - \left(\frac{\cot y - 1}{\cot y + 1} \right)} \right)$$

$$= \lim_{y \rightarrow 0} \left(\frac{\log \left(1 + \left(\frac{\cot y - 1}{\cot y + 1} - 1 \right) \right)}{- \left(1 - \left(\frac{\cot y - 1}{\cot y + 1} \right) \right)} \right)$$

$$= -1$$

89. Limit = $\lim_{x \rightarrow 0} \left(\frac{e^x - 1 - x}{x^2} \right)$

$$= \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{2x} \right)$$

$$= \frac{1}{2}$$

90. Limit = $\lim_{x \rightarrow 0} \left(\frac{e^{x^3} - 1 - x^3}{64x^6} \right)$

$$= \lim_{y \rightarrow 0} \left(\frac{e^y - 1 - y}{64 \cdot (y^2)} \right)$$

$$= \lim_{y \rightarrow 0} \frac{1}{64} \left(\frac{e^y - 1 - y}{(y^2)} \right)$$

$$= \lim_{y \rightarrow 0} \frac{1}{64} \left(\frac{e^y - 1}{2y} \right)$$

$$= \frac{1}{64} \times \frac{1}{2}$$

$$= \frac{1}{128}$$

$$\begin{aligned}
 91. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{e^x - e^{-x} - 2x}{x - \sin x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^x + e^{-x} - 2}{1 - \cos x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^x - e^{-x}}{\sin x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^x + e^{-x}}{\cos x} \right) \\
 &= 2
 \end{aligned}$$

$$\begin{aligned}
 92. \text{ Limit} &= \lim_{x \rightarrow e} \left(\frac{\ln x - 1}{x - e} \right) \\
 &= \lim_{x \rightarrow e} \left(\frac{\frac{1}{x}}{1} \right) \\
 &= \frac{1}{e}
 \end{aligned}$$

$$\begin{aligned}
 93. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{x + \tan 2x}{x - \tan 2x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{1 + \frac{\tan 2x}{x}}{1 - \frac{\tan 2x}{x}} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{1 + 2 \cdot \frac{\tan 2x}{2x}}{1 - 2 \cdot \frac{\tan 2x}{2x}} \right) \\
 &= \left(\frac{1 + 2}{1 - 2} \right) = -3
 \end{aligned}$$

$$\begin{aligned}
 94. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{\sin x - x}{x^3} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\cos x - 1}{3x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{-\sin x}{6x} \right) \\
 &= -\frac{1}{6}
 \end{aligned}$$

$$\begin{aligned}
 95. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{e^x + e^{-x} - x^2 - 2}{\sin^2 x - x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^x - e^{-x} - 2x}{\sin 2x - 2x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^x + e^{-x} - 2x}{2 \cos 2x - 2} \right)
 \end{aligned}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \left(\frac{e^x - e^{-x}}{-4 \sin 2x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^x + e^{-x}}{-8 \cos 2x} \right) \\
 &= -\frac{2}{8} = -\frac{1}{4}
 \end{aligned}$$

$$\begin{aligned}
 96. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{x e^x - \log(1+x)}{x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^x + x e^x - \frac{1}{1+x}}{2x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^x + e^x + x e^x + \frac{1}{(1+x)^2}}{2} \right) \\
 &= \frac{3}{2}
 \end{aligned}$$

$$\begin{aligned}
 97. \text{ Limit} &= \lim_{x \rightarrow \infty} \left(\frac{\ln x}{x} \right) \\
 &= \lim_{x \rightarrow \infty} \left(\frac{\frac{1}{x}}{1} \right) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 98. \text{ Limit} &= \lim_{x \rightarrow \infty} \left(\frac{x}{e^x} \right) \\
 &= \lim_{x \rightarrow \infty} \left(\frac{1}{e^x} \right) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 99. \text{ Limit} &= \lim_{x \rightarrow \infty} \left(\frac{x^{2014}}{e^x} \right) \\
 &= \lim_{x \rightarrow \infty} \left(\frac{2014 \cdot 2013 \dots 3 \cdot 2 \cdot 1}{e^x} \right) \\
 &= \lim_{x \rightarrow \infty} \left(\frac{(2014)!}{e^x} \right) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 100. \text{ Limit} &= \lim_{x \rightarrow 0} \left(\frac{x^2 + 2 \cos x - 2}{x \sin^3 x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{x^2 + 2 \cos x - 2}{x^4} \times \frac{x^3}{\sin^3 x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{x^2 + 2 \cos x - 2}{x^4} \right)
 \end{aligned}$$

$$= \lim_{x \rightarrow 0} \left(\frac{2x - 2\sin x}{4x^3} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{2 - 2\cos x}{12x^2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{2\sin x}{24x} \right)$$

$$= \frac{1}{12}$$

$$101. \text{ Limit} = \lim_{x \rightarrow \frac{\pi}{2}} (\tan x \log_e \sin x)$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\log_e \sin x}{\cot x} \right)$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\cot x}{-\operatorname{cosec}^2 x} \right)$$

$$= 0$$

$$102. \text{ Limit} = \lim_{x \rightarrow \infty} \left(\frac{x+6}{x+1} \right)^{x+4}$$

$$= \lim_{x \rightarrow \infty} \left(1 + \frac{5}{x+1} \right)^{\frac{x+1}{5} \times \frac{x+4}{x+1} \times 5}$$

$$= e^{\lim_{x \rightarrow \infty} \frac{1+\frac{4}{x}}{1+\frac{1}{x}} \times 5}$$

$$= e^5$$

$$103. \text{ Limit} = \lim_{x \rightarrow \infty} \left(\frac{2x+4}{2x+3} \right)^{2x+10}$$

$$= \lim_{y \rightarrow \infty} \left(\frac{y+4}{y+3} \right)^{y+10}$$

$$= \lim_{y \rightarrow \infty} \left(1 + \frac{1}{y+3} \right)^{\frac{y+3}{1} \times \frac{y+10}{y+3}}$$

$$= e^{\lim_{y \rightarrow \infty} \frac{y+10}{y+3}}$$

$$= e$$

$$104. \text{ Limit} = \lim_{x \rightarrow \infty} \left(\sin\left(\frac{1}{x}\right) + \cos\left(\frac{1}{x}\right) \right)^x$$

$$= \lim_{y \rightarrow 0} (\sin y + \cos y)^{\frac{1}{y}}$$

$$= \lim_{y \rightarrow 0} (1 + (\sin y + \cos y - 1))^{\frac{1}{y}}$$

$$= e^{\lim_{y \rightarrow 0} ((\sin y + \cos y - 1)) \times \frac{1}{y}}$$

$$= e^{\lim_{y \rightarrow 0} \left(\frac{(\sin y + \cos y - 1)}{y} \right)}$$

$$= e^{\lim_{y \rightarrow 0} ((\cos y - \sin y))}$$

$$= e$$

$$105. \text{ Limit} = \lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x}{3} \right)^{\frac{2}{x}}$$

$$= \lim_{x \rightarrow 0} \left(1 + \left(\frac{a^x + b^x + c^x}{3} - 1 \right) \right)^{\frac{2}{x}}$$

$$= \lim_{x \rightarrow 0} \left(1 + \left(\frac{a^x + b^x + c^x - 3}{3} \right) \right)^{\frac{2}{x}}$$

$$= e^{\lim_{x \rightarrow 0} \left(\left(\frac{a^x + b^x + c^x - 3}{3} \right) \times \frac{2}{x} \right)}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{(a^x - 1) + (b^x - 1) + (c^x - 1)}{x} \times \frac{2}{3} \right)}$$

$$= e^{\left(\frac{2}{3} \log(abc) \right)}$$

$$= (abc)^{2/3}$$

$$106. \text{ Limit} = \lim_{x \rightarrow 1} \left(2 - \frac{1}{x} \right)^{\tan\left(\frac{\pi x}{2}\right)}$$

$$= \lim_{y \rightarrow 0} \left(2 - \frac{1}{1+y} \right)^{\tan\left((1+y)\frac{\pi}{2}\right)}$$

$$= \lim_{y \rightarrow 0} \left(2 - \frac{1}{1+y} \right)^{-\cot\left(\frac{\pi y}{2}\right)}$$

$$= \lim_{y \rightarrow 0} \left(1 + \left(1 - \frac{1}{1+y} \right) \right)^{-\cot\left(\frac{\pi y}{2}\right)}$$

$$= \lim_{y \rightarrow 0} \left(1 + \left(\frac{y}{1+y} \right) \right)^{-\cot\left(\frac{\pi y}{2}\right)}$$

$$= e^{\lim_{y \rightarrow 0} \left(\frac{y}{1+y} \right) \times -\cot\left(\frac{\pi y}{2}\right)}$$

$$= e^{\lim_{y \rightarrow 0} \left(\frac{-1}{1+y} \right) \times \frac{\left(\frac{\pi y}{2}\right)}{\tan\left(\frac{\pi y}{2}\right)} \times \frac{2}{\pi}}$$

$$= e^{-\frac{2}{\pi}}$$

$$107. \text{ Limit} = \lim_{x \rightarrow 1} \left(\tan\left(\frac{\pi}{4} + \ln x\right) \right)^{\frac{1}{\ln x}}$$

$$= \lim_{x \rightarrow 1} \left(\frac{1 + \tan(\log x)}{1 - \tan(\log x)} \right)^{\frac{1}{\log x}}$$

$$= \lim_{x \rightarrow 1} \left(1 + \frac{2\tan(\log x)}{1 - \tan(\log x)} \right)^{\frac{1}{\log x}}$$

$$= \lim_{x \rightarrow 1} \left(\frac{2 \tan(\log x)}{1 - \tan(\log x)} \right) \times \frac{1}{\log x}$$

$$= e^2$$

108. Limit = $\lim_{x \rightarrow 0} \left(\frac{\ln \left(\tan \left(\frac{\pi}{4} + 2x \right) \right)}{\sin 3x} \right)$

$$= \lim_{x \rightarrow 0} \left(\frac{\log \left(\frac{1 + \tan 2x}{1 - \tan 2x} \right)}{\sin 3x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\log \left(1 + \left(\frac{1 + \tan 2x}{1 - \tan 2x} - 1 \right) \right)}{\sin 3x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\log \left(1 + \left(\frac{2 \tan 2x}{1 - \tan 2x} \right) \right)}{\left(\frac{2 \tan 2x}{1 - \tan 2x} \right)} \times \frac{\left(\frac{2 \tan 2x}{1 - \tan 2x} \right)}{\sin 3x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\left(\frac{2 \tan 2x}{1 - \tan 2x} \right)}{\sin 3x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\left(\frac{\tan 2x}{2x} \right) \times \frac{3x}{\sin 3x} \times \frac{2x}{3x} \times 2 \right)$$

$$= \frac{4}{3}$$

109. Limit = $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x}$

$$= \lim_{x \rightarrow 0} \left(1 + \left(\frac{\tan x}{x} - 1 \right) \right)^{1/x}$$

$$= \lim_{x \rightarrow 0} \left(1 + \left(\frac{\tan x - x}{x} \right) \right)^{1/x}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{\tan x - x}{x} \right) \times \frac{1}{x}}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{\tan x - x}{x^2} \right)}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{\sec^2 x - 1}{2x} \right)}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{2 \sec^2 x \tan x}{2} \right)}$$

$$= 1$$

110. Limit = $\lim_{x \rightarrow 0} \left(\frac{1 + \tan x}{1 + \sin x} \right)^{\operatorname{cosec} x}$

$$= \lim_{x \rightarrow 0} \left(1 + \left(\frac{1 + \tan x}{1 + \sin x} - 1 \right) \right)^{\operatorname{cosec} x}$$

$$= \lim_{x \rightarrow 0} \left(1 + \left(\frac{\tan x - \sin x}{1 + \sin x} - x \right) \right)^{\operatorname{cosec} x}$$

$$= e^{\lim_{x \rightarrow 0} \left(\left(\frac{\tan x - \sin x}{1 + \sin x} \right) \times \frac{1}{\sin x} \right)}$$

$$= e^{\lim_{x \rightarrow 0} \left(\sin x \left(\frac{\sec x - 1}{1 + \sin x} \right) \times \frac{1}{\sin x} \right)}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{\sec x - 1}{1 + \sin x} \right)}$$

$$= 1$$

111. Limit = $\lim_{n \rightarrow \infty} \left(\frac{a - 1 + \sqrt[n]{b}}{a} \right)^n$

$$= \lim_{n \rightarrow \infty} \left(1 + \left(\frac{\sqrt[n]{b} - 1}{a} \right) \right)^n$$

$$= e^{\lim_{n \rightarrow \infty} \left(\left(\frac{\sqrt[n]{b} - 1}{a} \right) \times n \right)}$$

$$= e^{\lim_{n \rightarrow \infty} \left(\left(\frac{b^{\frac{1}{n}} - 1}{a} \right) \times n \right)}$$

$$= e^{\lim_{n \rightarrow \infty} \left(\frac{b^{\frac{1}{n}} - 1}{\frac{1}{n}} \right) \times \frac{1}{a}}$$

$$= e^{\frac{1}{a} \log b}$$

$$= e^{\log b^{\frac{1}{a}}}$$

$$= b^{\frac{1}{a}}$$

112. We have, $-1 \leq \sin \left(\frac{1}{x} \right) \leq 1$

$$\Rightarrow -x^2 \leq x^2 \sin \left(\frac{1}{x} \right) \leq x^2$$

Now, $\lim_{x \rightarrow 0} (x^2) = 0$

and $\lim_{x \rightarrow 0} (-x^2) = 0$

Thus, $\lim_{x \rightarrow 0} \left(x^2 \sin \left(\frac{1}{x} \right) \right) = 0$

113. We have, $-1 \leq \sin x \leq 1$

$$\Rightarrow -\frac{1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$$

Now, $\lim_{x \rightarrow \infty} \left(-\frac{1}{x} \right) = 0$

and $\lim_{x \rightarrow \infty} \left(\frac{1}{x} \right) = 0$

Thus, $\lim_{x \rightarrow \infty} \left(\frac{\sin x}{x} \right) = 0$

114. We have, $x - 1 < [x] \leq x$

$$\Rightarrow \frac{x-1}{x} < \frac{[x]}{x} \leq \frac{x}{x}$$

$$\Rightarrow \left(1 - \frac{1}{x}\right) < \frac{[x]}{x} \leq 1$$

$$\text{Now, } \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right) = 1$$

$$\text{and } \lim_{x \rightarrow \infty} (1) = 1$$

$$\text{Thus, } \lim_{x \rightarrow \infty} \left(\frac{[x]}{x}\right) = 1$$

115. We have, $x - 1 < [x] \leq x$

$$\Rightarrow \frac{1}{x} \leq \frac{1}{[x]} < \frac{1}{x-1}$$

$$\Rightarrow \frac{x}{x} \leq \frac{x}{[x]} < \frac{x}{x-1}$$

$$\Rightarrow 1 \leq \frac{x}{[x]} < \frac{x}{x-1}$$

$$\text{Now, } \lim_{x \rightarrow \infty} (1) = 1$$

$$\text{and } \lim_{x \rightarrow \infty} \left(\frac{x}{x-1}\right) = 1$$

$$\text{Thus, } \lim_{x \rightarrow \infty} \left(\frac{x}{[x]}\right) = 1$$

116. We have, $x - 1 < [x] \leq x$

$$2x - 1 < [2x] \leq 2x$$

$$3x - 1 < [3x] \leq 3x$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$nx - 1 < [nx] \leq nx$$

By adding we get,

$$(1 + 2 + 3 + \dots + n)x - n < ([x] + [2x] + [3x] + \dots + [nx]) \leq (1 + 2 + 3 + \dots + n)x - n$$

$$\Rightarrow \frac{(1 + 2 + 3 + \dots + n)x - n}{n^2} < \frac{([x] + [2x] + [3x] + \dots + [nx])}{n^2} \leq \frac{(1 + 2 + 3 + \dots + n)x}{n^2}$$

$$\text{Now, } \lim_{x \rightarrow \infty} \frac{(1 + 2 + 3 + \dots + n)x - n}{n^2}$$

$$= \lim_{x \rightarrow \infty} \left(\frac{n(n+1)x}{2n^2} - \frac{1}{n} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\left(1 + \frac{1}{n}\right) \frac{x}{2} - \frac{1}{n} \right)$$

$$= \frac{x}{2}$$

$$\text{and } \lim_{x \rightarrow \infty} \frac{(1 + 2 + 3 + \dots + n)x}{n^2}$$

$$= \lim_{x \rightarrow \infty} \left(\frac{n(n+1)x}{2n^2} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\left(1 + \frac{1}{n}\right) \frac{x}{2} \right)$$

$$= \frac{x}{2}$$

$$\text{Thus, } \lim_{x \rightarrow \infty} \left(\frac{[x] + [2x] + [3x] + \dots + [nx]}{n^2} \right) = \frac{x}{2}$$

$$117. \text{ Limit} = \lim_{x \rightarrow \infty} \left(\frac{x - \int_0^x \cos(t^2) dt}{x^3 - 6x} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\frac{1 - \cos(x^2)}{3x^2 - 6} \right)$$

$$= \left(\frac{1 - 1}{0 - 6} \right)$$

$$= 0$$

$$118. \text{ Limit} = \lim_{x \rightarrow \infty} \left(\frac{\int_0^x \sqrt{4 + t^4} dt}{x^3} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\frac{\sqrt{4 + x^4}}{3x^2} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\sqrt{\frac{4 + x^4}{9x^4}} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\sqrt{\frac{4}{9x^4} + \frac{1}{9}} \right)$$

$$= \frac{1}{3}$$

$$119. \text{ Limit} = \lim_{x \rightarrow 1} \left(\frac{\int_x^{x^2} e^{-t^2} dt}{x - 1} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{e^{-x^4} \cdot 2x - e^{-x} \cdot 1}{1} \right)$$

$$= -1$$

$$\begin{aligned}
120. \text{ Limit} &= \lim_{n \rightarrow \infty} \left[\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n} \right] \\
&= \lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{n+n} \right) \\
&= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{n}{n+r} \right) \\
&= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{1}{1 + \left(\frac{r}{n}\right)} \right) \\
&= \int_0^1 \frac{dt}{1+x} \\
&= \log 2
\end{aligned}$$

$$\begin{aligned}
121. \text{ Limit} &= \lim_{n \rightarrow \infty} \left[\frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \frac{n}{n^2+3^2} + \dots + \frac{1}{2n} \right] \\
&= \lim_{n \rightarrow \infty} \left[\frac{n}{n^2+1^2} + \frac{n}{n^2+1^2} + \dots + \frac{n}{n^2+n^2} \right] \\
&= \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{n}{n^2+r^2} \right) \\
&= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{n^2}{n^2+r^2} \right) \\
&= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{1}{1 + \left(\frac{r}{n}\right)^2} \right) \\
&= \int_0^1 \frac{dt}{1+x^2} \\
&= (\tan^{-1}x)|_0^1 \\
&= (\tan^{-1}(1) - \tan^{-1}(0)) \\
&= \frac{\pi}{4}
\end{aligned}$$

$$\begin{aligned}
122. \text{ Limit} &= \lim_{n \rightarrow \infty} \sum_{r=1}^{2n} \left(\frac{r}{\sqrt{n^2+r^2}} \right) \\
&= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \left(\frac{nr}{\sqrt{n^2+r^2}} \right) \\
&= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \left(\frac{\frac{r}{n}}{\sqrt{1 + \left(\frac{r}{n}\right)^2}} \right) \\
&= \int_0^2 \left(\frac{x}{\sqrt{1+x^2}} \right) dx \\
&= (\sqrt{1+x^2})|_0^2 \\
&= (\sqrt{5}-1)
\end{aligned}$$

$$\begin{aligned}
123. \text{ Limit} &= \lim_{n \rightarrow \infty} \sum_{r=1}^{2n} \frac{2}{n} \left(\frac{2r}{n} + 1 \right) \\
&= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} 2 \left(\frac{r}{n} \right) \left(2 \frac{r}{n} + 1 \right) \\
&= \int_0^2 2x(2x+1) dx \\
&= \int_0^2 (4x^2 + 2x) dx \\
&= \left(\frac{4x^3}{3} + x^2 \right) \Big|_0^2 \\
&= \left(\frac{16}{3} + 4 \right) = \frac{28}{3}
\end{aligned}$$

$$\begin{aligned}
124. \text{ Let } A &= \lim_{n \rightarrow \infty} \left(\frac{(2n!)!}{n! \cdot n^n} \right)^{1/n} \\
&= \lim_{n \rightarrow \infty} \left(\frac{(2n(2n-1)(2n-2)\dots(2n-(n-1)))n!}{n!n^n} \right)^{1/n} \\
&= \lim_{n \rightarrow \infty} \left(\frac{(2n(2n-1)(2n-2)\dots(2n(n-1)))}{n^n} \right)^{1/n} \\
&\Rightarrow \text{Log } A \\
&= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{n-1} \log \left(\frac{2n-r}{n} \right) \\
&= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{n-1} \log \left(2 - \frac{r}{n} \right) \\
&= \int_0^1 \log(2-x) dx \\
&= -\int_2^1 \log y dy, \text{ where } (2-x) = y \\
&= (y - y \log y) \Big|_2^1 \\
&= 1 - 2 + 2 \log 2 \\
&= \log 4 - 1 \\
&= \log 4 - \log e \\
&= \log \left(\frac{4}{e} \right)
\end{aligned}$$

$$\text{Thus, } A = \frac{4}{e}$$

$$\begin{aligned}
125. \lim_{x \rightarrow 0^+} (x - x^2)^{x^2} &= e^{\lim_{x \rightarrow 0^+} x^2 \log(x - x^2)} \\
&= e^{\lim_{x \rightarrow 0^+} \left(\frac{x}{x-1} \right) ((x-x^2) \log(x-x^2))} \\
&= e^0 \quad (\because \lim_{x \rightarrow 0^+} x \log x = 0) = 1
\end{aligned}$$